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Customer Churn Analysis in Banking Institutions

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Variable names and figure labels are kept as in the original.

Abstract

This report addresses the prediction of customer churn in the banking sector, a key input for the effectiveness of retention strategies. Using data on bank customers, the goal was to build and compare two classifiers of different nature: a logistic regression model and a classification tree.

After data validation and cleaning, a training set (1,000 observations) and a test set (600 observations) were extracted. Given the class imbalance of the response variable, overall accuracy was rejected as the main metric, and performance was assessed mainly through sensitivity and the area under the ROC curve (AUC).

The results show that logistic regression outperforms the classification tree. Setting the threshold with the Youden index allowed the logistic model to reach a sensitivity of 87.40%, confirming this approach as the most suitable for the business context, since it maximises the identification of customers who actually leave.

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1 Introduction

Banks currently offer a wide range of services that go well beyond traditional banking activities. Customers increasingly look for solutions that fit their needs and often switch institutions when they are not satisfied.

In this context, it is essential for banks to regularly analyse their customer profiles, in order to identify opportunities for improvement and strengthen loyalty. In particular, it is crucial to understand which characteristics are associated with potential churn and, ideally, to develop models that anticipate it, making it possible to implement effective retention strategies.

The main objective of this report is to address this challenge by building and comparing the performance of classifiers for the occurrence of churn, based on a banking dataset. The focus is on identifying the model that best balances predictive accuracy and practical usefulness, providing adequate support for retention decisions.

2 Methodology

The methodological approach was organised in four sequential phases: data preprocessing, sample partition, construction of the classification algorithms, and definition of the performance metrics.

2.1 Preprocessing and Data Transformation

The first phase consisted of importing, validating and cleaning the original dataset. Observations with missing values were removed, keeping only complete cases. Categorical variables were converted to factors and continuous variables were kept as numeric.

During exploratory analysis, the categorical variable for the number of bank products (`NumProdutos`) was found to have very small samples in the categories of 3 and 4 products (22 and 3 observations, respectively). Because of these small sample sizes, and to avoid problems in parameter estimation, these categories were merged into a single class ("3+"), which replaced the original variable in the modelling process.

Table 1: Variables used (original names, as they appear in the figures and tables).

Variable	Meaning
`Abandono`	Churn (response: 1 = left the bank, 0 = stayed)
`Idade`	Age
`Saldo`	Account balance
`ScoreCredito`	Credit score
`Antiguidade`	Tenure (customer seniority)
`SalarioEstimado`	Estimated salary
`NumProdutos`	Number of bank products (3 and 4 merged into "3+")
`Geografia`	Country (France, Germany, Spain)
`Genero`	Gender
`CartaoCred`	Holds a credit card
`ClienteAtivo`	Active customer

2.2 Data Partition

To obtain an unbiased assessment of the generalisation ability of the predictive models, the data were split into two independent subsets: a training set ($n = 1000$) and a test set ($n = 600$). The

training set was used exclusively to estimate the algorithms, while the test set was kept strictly for performance evaluation.

To ensure full reproducibility of the random sampling, a different seed was fixed for each set.

2.3 Classification Algorithms

Using the training set, two predictive models were fitted, which allows two different statistical approaches to be compared, one of them a Machine Learning algorithm based on partitioning:

- **Logistic regression:** a model assuming a binomial distribution for the response variable, with the logit link function. Automatic variable selection (bidirectional stepwise) was applied. The final model was the one that minimised the Akaike Information Criterion (AIC).
- **Classification tree:** a model based on recursive binary partitions. Decision trees tend to overfit if grown to their maximum depth. To avoid this, the initial tree was pruned by cost-complexity pruning. The selected complexity parameter (CP) was the one that minimised the cross-validation error (`xerror`).

2.4 Performance Metrics

Because of the class imbalance in the response variable (`Abandono`), overall accuracy is inadequate and biased towards the majority class. Performance on the test set was therefore based on:

- **Sensitivity and specificity:** the ability of the model to correctly identify customers who leave (true positives) and its ability not to wrongly classify customers as churners (true negatives), respectively.
- **Area under the ROC curve (AUC):** the overall discriminatory ability of the classifier across all possible decision thresholds.

In addition, applying the logistic regression model directly requires a cut-off (threshold). Instead of the conventional threshold of 0.5, the Youden index (J) was used:

$$J = \text{Sensitivity} + \text{Specificity} - 1.$$

This identifies the cut-off that maximises the sum of sensitivity and specificity.

3 Analysis

3.1 Training Data

The training set has 1,000 observations, of which 810 (81.00%) are customers who stayed and 190 (19.00%) are customers who left.

3.2 Exploratory Data Analysis

3.2.1 Continuous Variables

Boxplots by churn group were produced for the continuous variables (Figure 1). `Idade` (age) stands out as the numeric predictor with the greatest discriminatory power: customers who

leave have a visibly higher median age than those who stay, suggesting that older customers are more likely to churn. The group that stays also has several outliers.

For `Saldo` (balance), higher values are also found among customers who leave. There may be a relationship between the two variables.

`ScoreCredito`, `Antiguidade` and `SalarioEstimado` have very similar distributions across the two groups, suggesting weak predictive ability. The correlation matrix (Figure 2) also confirms that the numeric variables are practically uncorrelated with each other, so no multicollinearity problems are expected when estimating the models.

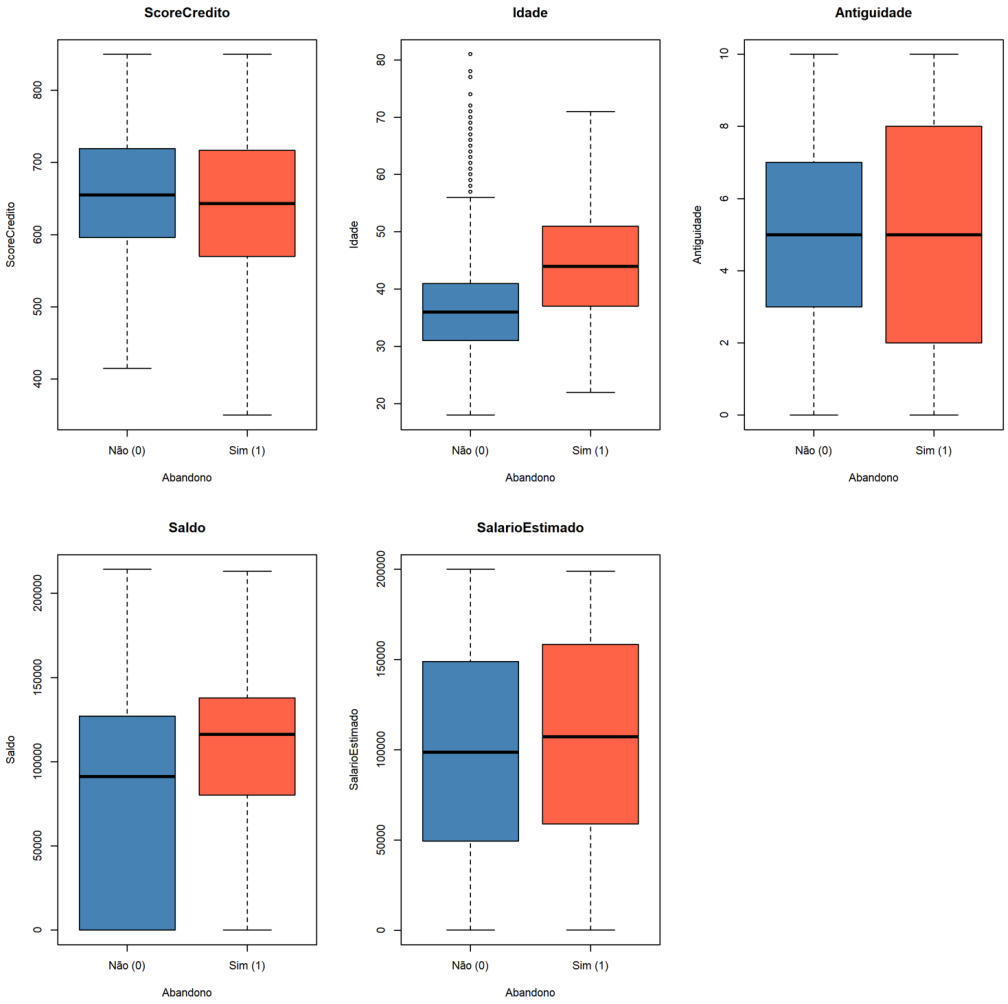


Figure 1: Boxplots of the continuous variables by churn group (training set). Axis labels in Portuguese: *Não (0)* = stayed, *Sim (1)* = left.

3.2.2 Categorical Variables

For the categorical variables, tables of the churn proportion by category were computed and stacked bar charts were produced (Figure 3), making it possible to identify the groups most prone to churn.

`NumProdutos` shows the most striking pattern: customers with 2 products have the lowest churn rate, while customers with only 1 product leave in about 28.00% of cases. The categories with more than 3 products, although they show high churn rates, have very small samples (25 observations) and should be interpreted with caution.

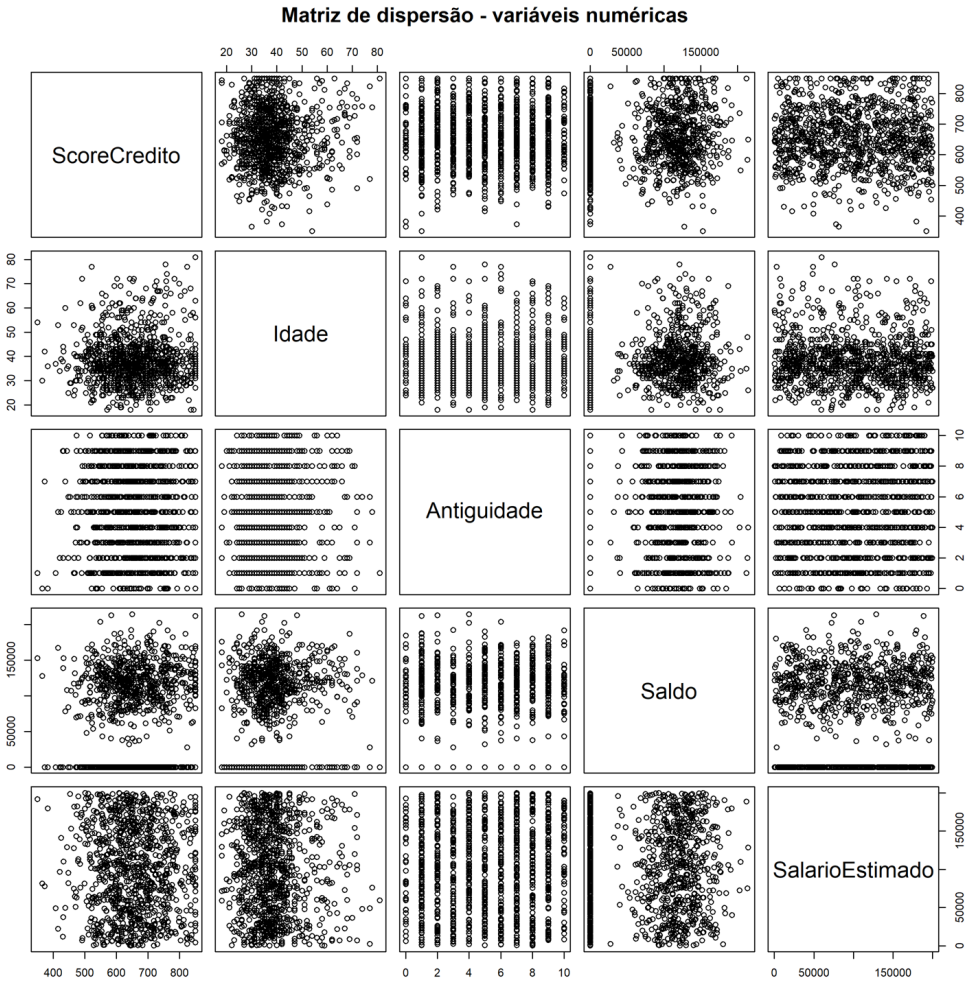


Figure 2: Scatter plot matrix of the numeric variables (training set).

Regarding `Geografia`, customers from Germany have a higher churn rate than French and Spanish customers, suggesting that location is a relevant factor.

`ClienteAtivo` shows that inactive customers leave the institution more often than active ones, which is logical and easy to interpret. Inactivity may be a warning sign of churn.

`Genero` shows a smaller difference, with female customers having a slightly higher churn rate than male customers.

Finally, `CartaoCred` shows no significant differences between its categories, suggesting that holding a credit card has little association with churn.

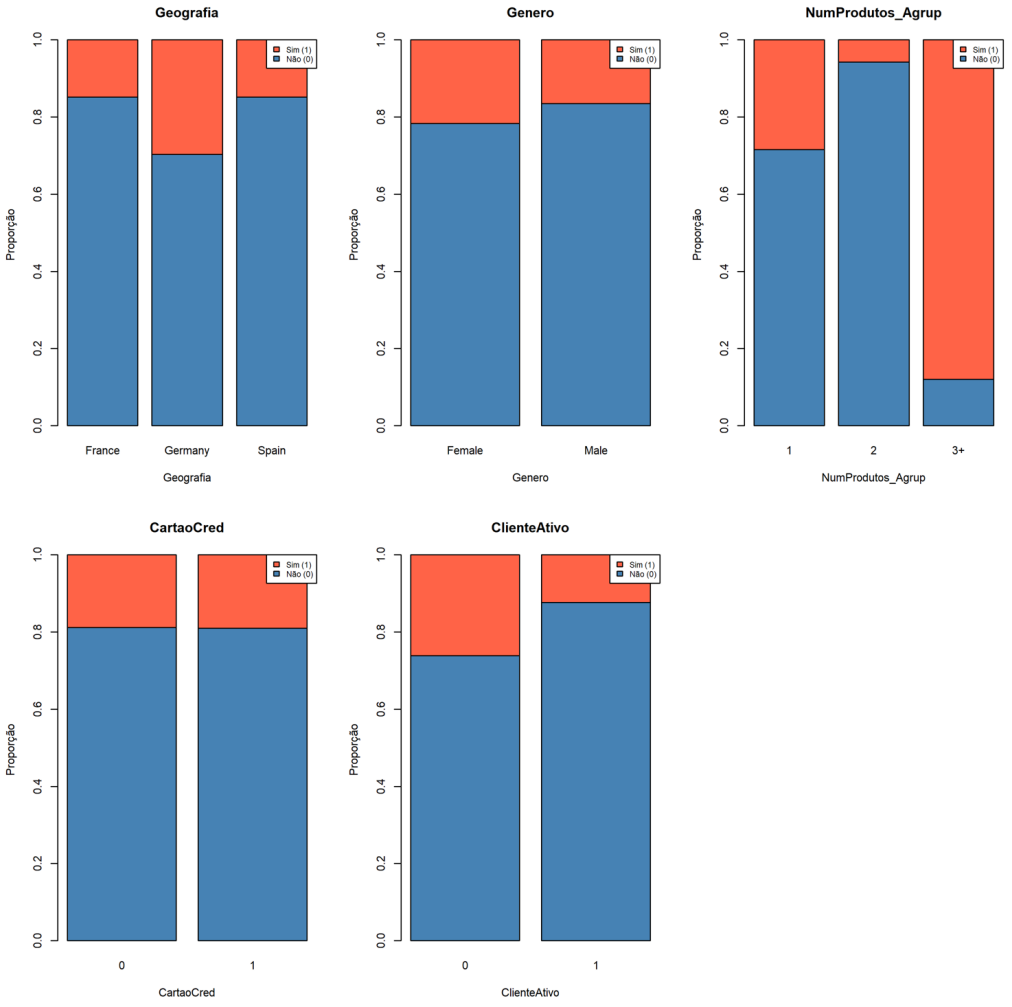


Figure 3: Proportion of churn by category for each categorical variable (training set). In the legend, *Sim (1)* = left and *Não (0)* = stayed.

3.3 Test Data

The classifiers were evaluated on the test set of 600 observations, of which 473 (78.83%) are customers who stayed and 127 (21.17%) are customers who left.

3.4 Logistic Regression

The variables retained by the stepwise algorithm were `NumProdutos`, `Idade`, `ClienteAtivo`, `Geografia`, `Genero`, `SalarioEstimado` and `ScoreCredito`. `NumProdutos`, `Idade`, `ClienteAtivo`, `Geografia` and `Genero` are the most significant. Although the other two are not statistically significant, they remain in the model because they carry relevant information under the AIC criterion.

For the logistic regression, the classification threshold was set with the Youden index, which maximises the sum of sensitivity and specificity. This is more appropriate than the conventional 0.5 threshold in a context where detecting churn is the priority.

With the Youden threshold (0.15), the model reaches a sensitivity of 87.40%, correctly identifying 111 of the 127 customers who left. This is the most relevant result from a business point of view.

The specificity of 64.50% and the accuracy of 69.30% are lower, which reflects the trade-off inherent to the Youden index: the model becomes more conservative in classifying "does not leave" and produces more false positives. The AUC of 0.83 confirms good overall discriminatory ability (Figure 4).

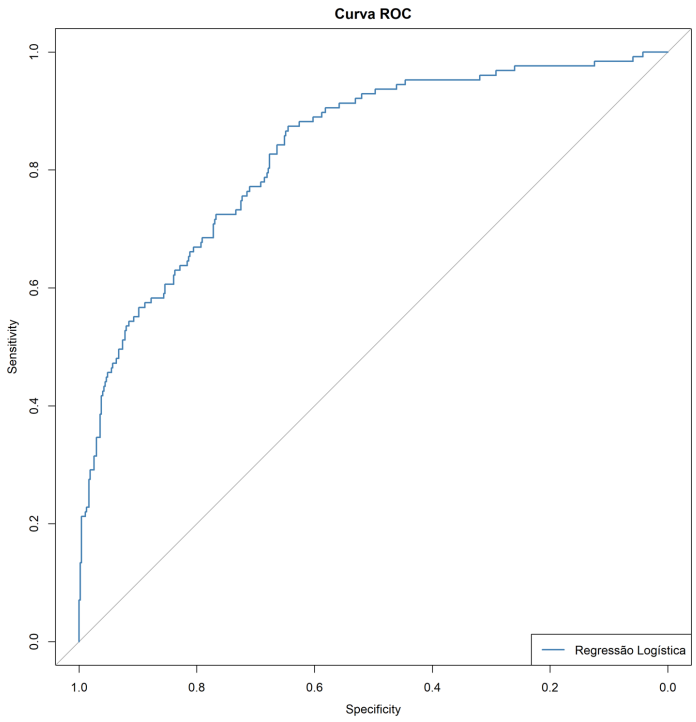


Figure 4: ROC curve of the logistic regression model on the test set (AUC = 0.83).

3.5 Classification Tree

The tree has an accuracy of 80.80% and a high specificity (94.10%), but a sensitivity of only 31.50%: it correctly identifies only 40 of the 127 customers who left.

This profile is clearly inadequate for a customer retention problem, where the central goal is precisely to anticipate who will leave. The AUC of 0.73 confirms a lower overall discriminatory ability than the logistic regression (Figure 5).

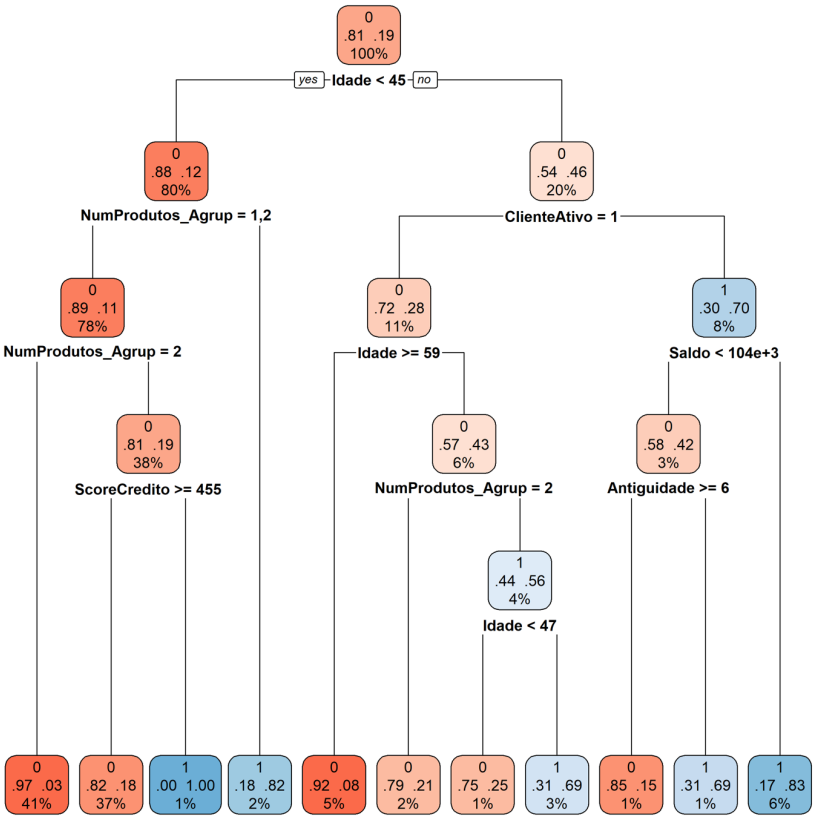


Figure 5: Classification tree, with class proportions and percentage of observations in each node.

3.6 Comparative Analysis

The two models show opposite performance profiles. The tree has higher accuracy and specificity, but the logistic regression outperforms it on the metrics that matter for this problem, sensitivity and AUC. The difference in AUC (0.83 vs. 0.73) is the most telling, because it is independent of the threshold and reflects the overall discriminatory ability of each model, as confirmed visually by the ROC curves (Figure 6).

Because it produces a limited number of distinct thresholds, the tree has less flexibility in the sensitivity/specificity trade-off, which shows in its ROC curve being closer to the diagonal. By modelling continuous probabilities directly, the logistic regression allows this trade-off to be adjusted better.

Table 2: Performance of the classifiers on the test set.

Model	Accuracy	Sensitivity	Specificity	AUC
Logistic regression (Youden)	69.30%	87.40%	64.50%	0.83
Classification tree	80.80%	31.50%	94.10%	0.73

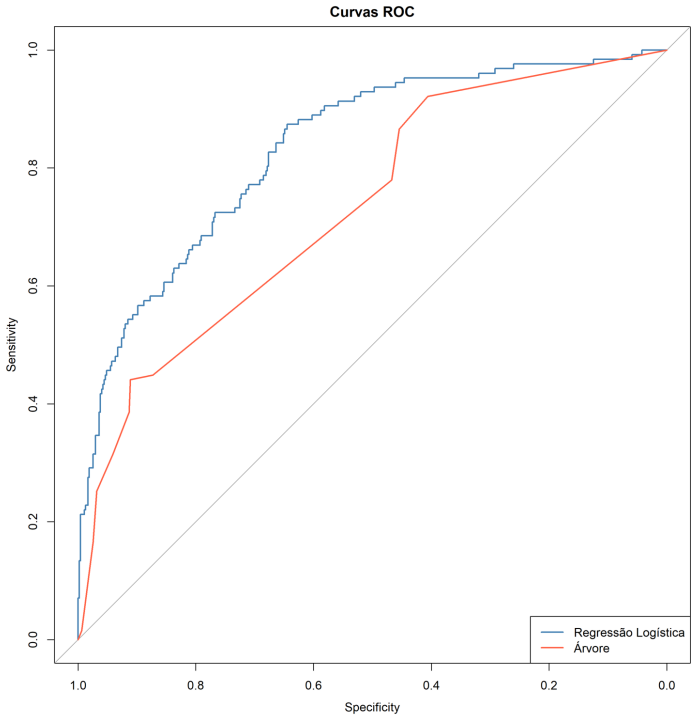


Figure 6: Overlaid ROC curves of the two classifiers on the test set (blue: logistic regression, red: classification tree).

4 Conclusions and Discussion

The variables identified in the exploratory analysis as having the greatest discriminatory power, namely `Idade`, `NumProdutos`, `Geografia` and `ClienteAtivo`, were retained by the stepwise algorithm, which reinforces the need for a good exploratory analysis. The only exception was `Saldo`.

Logistic regression with the Youden threshold proved to be the most suitable classifier for the

goal of customer retention, correctly identifying 87.40% of the customers who actually left. The trade-off was a specificity of 64.50%, which means more false positives. This is acceptable in the business context, because the cost of failing to identify a real churner tends to be higher than the cost of unnecessarily applying a retention strategy to a customer.

The classification tree, despite its higher accuracy (80.80%), proved inadequate for this problem, correctly identifying only 31.50% of the churners. A classifier that fails to anticipate more than two thirds of the customers who leave has very limited practical usefulness, regardless of its accuracy.

The main limitation is that the models were built and evaluated on a single partition of the data, which may introduce variability in the results. In future work it would be worthwhile to explore techniques such as SMOTE and/or resampling to deal with class imbalance, or more complex models such as random forests.

Note. The AI model Gemini (Google) was used as a study-support tool for this project.

References

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